FACULTY OF ENGINEERING AND ARCHITECTURE

DETOUR

Technical report

Course: Design Thinking

Students: Team 07

Jan Kino	Jan.Kino@UGent.be
Kevin Deneut	Kevin.Deneut@UGent.be
Sam Decorte	Sam.Decorte@UGent.be
Senne Van Nieuwenhuyse	Senne.VanNieuwenhuyse@UGent.be

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GHENT UNIVERSITY

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1 Introdution

this is the introduction

2 Design thinking phases

Our design thinking phase was far from a linear process. Some phases were passed through multiple times/revisited. In the following sections, the different phases will be discussed as one complete phase, which for us doesn't mean that all the described actions, decisions, etc. were made in the same week/period.

2.1 Empathize phase

In our empathize phase we decided on what our stakeholders and target personas would be. We went through this process mainly by first having our initial meeting with our challenger Geert Crombez. After this meeting our whole team felt like we didn't know what direction we could/should go in. Then we had a team meeting and decided upon our target group for behaviour change, namely smokers. By making the decision to make a smoke-stop app, we then looked for (ex-)smokers to interview and see/feel what the needs of those people are in the process to stop smoking (by using an app). Out of those interviews, 2 main themes came back each time:

1. The path to stop smoking has to start and come from within the person themselves.
2. In the process of stopping to smoke, rational arguments (e.g. health benefits or saved money) don't really make a difference.

The first theme was also something our challenger already mentioned in our first meeting, so it became clear to us that our app would not have to convince people who don't want to quit, but help the people who already do.

The (ex-)smokers we met could be divided into three groups: people who tried to stop before and failed, people who have stopped and want to keep it that way, and people who are trying to stop for the first time. Next to the 2 themes above, the interviews also showed us two needs they have. First, smokers don't know which method or app suits them. There are a lot of apps, but most of them are of low quality, and testing them one by one costs time and motivation. On top of that, a failed attempt with the wrong app feels like a personal failure. Second, cravings can have very different causes (e.g. stress, a certain place, boredom, a social situation or physical withdrawal), and the same technique doesn't work for every cause.

Although the challenge seemed pretty difficult at this stage, we decided to keep going and chose (ex-)smokers as end-users for our app.

The thing we liked most about this phase was that we felt like there were still a lot of paths we could follow and directions we could explore. Because the interviews showed us that rational

arguments don't really work, we knew we had to look further than the typical smoke-stop apps, which gave us a lot of freedom for the next phases.

2.2 Define phase

The main event that happened and shaped our define phase was our talk with the PhD student Arian. He worked on research about using chatbots (with direct interventions) to help people stop smoking by detecting craving moments. Although we now know that the outcome of their research is that sometimes getting reminded of your cravings leads to more smoking, the meeting gave us a great boost in motivation and gave us a clear vision to keep going and put our heads in the same direction. Something else we learned from this meeting is that we would be needing a thorough SOTA to see what already has been done in this space and where our app/method could go. Now we were ready to define our HMW question. To do this, we clustered all the HMW questions we came up with. The two largest clusters were "the right method for the right person" and "help at the moment of the craving", which match the two needs we found in our interviews. Taking the meeting with Arian into account, we decided we wanted to touch on a wider range of interventions to help people stop smoking, and not only work with a chatbot. So our define phase concluded with the following HMW question: "How might we choose the correct (direct vs. indirect) kind of intervention (BCT) for the right person at the correct moment (in the day) to stop smoking?" The clear thing we all liked about this phase of our design thinking process is that for the first time, we felt like all our heads were in the same direction. We also could all bounce ideas off each other way more easily.

2.3 Ideate phase

With our HMW question in mind, we started our ideate phase by looking at all the different ways we could help people stop smoking. Quite fast, we realised that there might be a lot of things that are possible, and that it would be impossible to work out all of them. Taking into account what we learned from the interviews and from our meeting with Arian, we decided that a combination of direct and indirect interventions would be the way to go. Direct interventions alone (e.g. a chatbot reacting on a craving moment) can sometimes do more harm than good, while indirect interventions alone might not be strong enough when someone really needs help. By combining both, we could try to choose the right kind of intervention for the right person at the right moment, which is exactly what our HMW question asks for. During this whole process, we always kept GDPR and privacy in mind, since our app would be working with very personal information.

Something we learned in this phase is that there are a lot of ways to get to our goal. There are lots of media that could be used to gather personal info about a person (e.g. their phone usage, location or wearables), which could help us to detect the right moment for an intervention. But maybe not always should we use all of them, because not everything that is technically possible is also allowed or desirable when it comes to the privacy of our users. So for every idea, we not only had to ask ourselves if it would work, but also if it would be okay for the

user that we use this information.

The thing we liked most about this phase was that we could explore all the options that might be available to get to our goal. Because we didn't have to choose right away, everyone could bring in their own ideas, no matter how crazy they seemed at first, which led to a lot of interesting discussions within our team.

2.4 First and second prototype

To bring our idea into a testable form, we created a prototype consisting of two different parts: a storyboard that would give an outline of how a possible user might interact with the app, and a user interface, to be able to present what the actual concept might look like for an end user.

2.4.1 Storyboard

Below is a step by step explanation of the flow of interaction with the app.

1. A smoker comes in contact with the app.
2. After downloading the app, the user will fill in a questionnaire with imporant information, through a conversation with the chatbot.
3. Scenario: the user is trying to stop from this day onwards (which we learned from the questionnaire).
4. On day 1, at a certain time, using wearable and other provided data by the user, the app detects high risk of craving and sends a notification, for example, a message from the chatbot.
5. The user might then actually interact with the notification, or ignore it.
6. Based on the amount of interaction with the user, and possibly some new questions asked by the chatbot, the app tries to assess wether the user actually smoked, and if the notification was effective.
7. If the notification was not effective, the next day, or at the next high risk moment, the app will try to send a different notification, with a different message, or at a different time. This could now be a notification for a funny news story, that might temporarily distract the user from their craving.
8. Again, after re-assessing the success. The app learns that this intervention was effective or not.
9. If these notifications keep failing, the craving support notifications might not be the right method for this specific user.

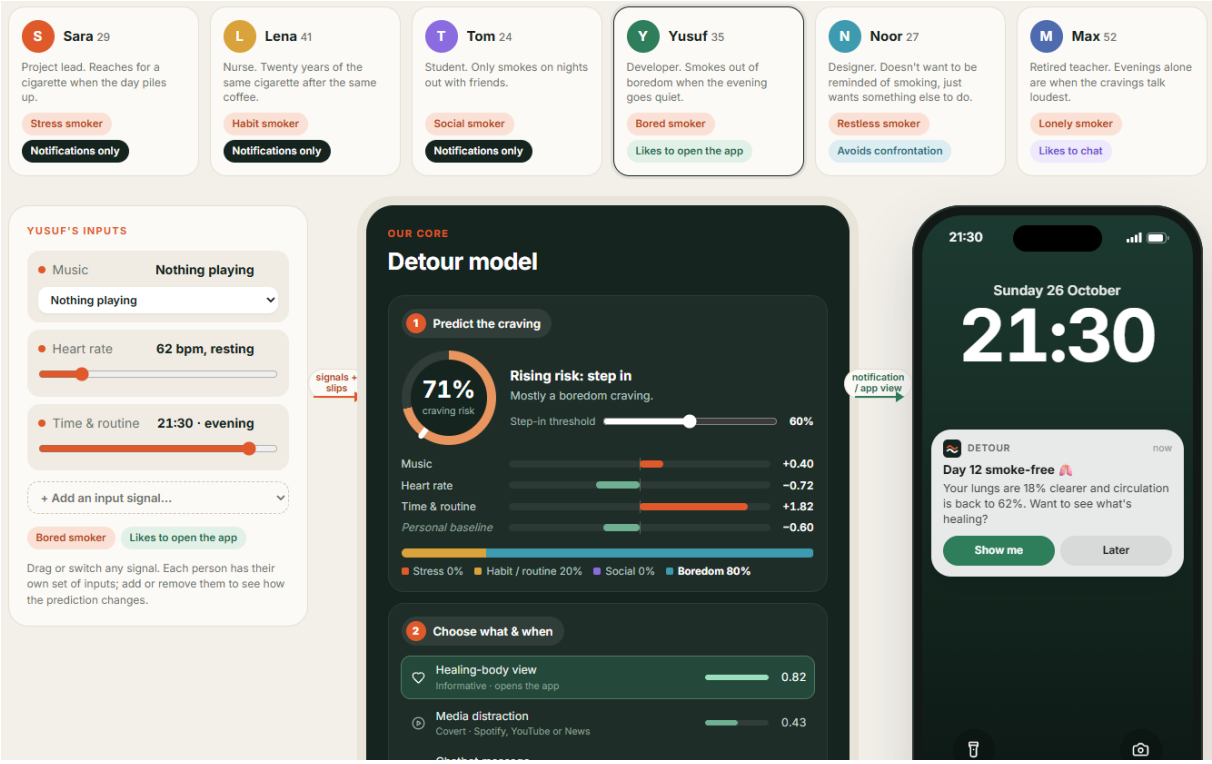


Figure 1: The visual prototype, with different example users (top), possible input data (bottom left), intervention and feature selection model (bottom center), visualisation of the user phone (bottom right).

10. The app should then learn this, and try a different feature, such as: helping the user plan their day to be less stressfull, or provide them with plans to cope with craving moments without their phone.

As a summary, this storyboard higlights our idea of the smart smoking cessation app: It keeps learning about the user, to try and choose the right features at the right time, and adapting the specific feature to their preferences. It combines direct, smoking cessation support, with indirect support, such as stress management and distraction techniques. Depending on what works best for the user, and what they prefer.

2.4.2 User interface

The visual interface was built using Figma, and later translated to an interactive web-app using AI-assisted coding. Figure 1 shows a large view of the prototype. At the top, we have an option to select different possible users with their own preferences and needs. Below this, on the left side, we have sliders to change the instantaneous input data, such as stress, steps today, daily routine, location. In the middle, we see a visual representation of what the model should be doing under the hood: translate this input data to intervention selection, such as media distraction, chatting with a chatbot. This is better seen in the enlarged view of the last 2 steps in figure 2. Finally, on the right, we have what the user would actually see on their phone: a personalized notification (figure 1), leading them to one of the app features, or a

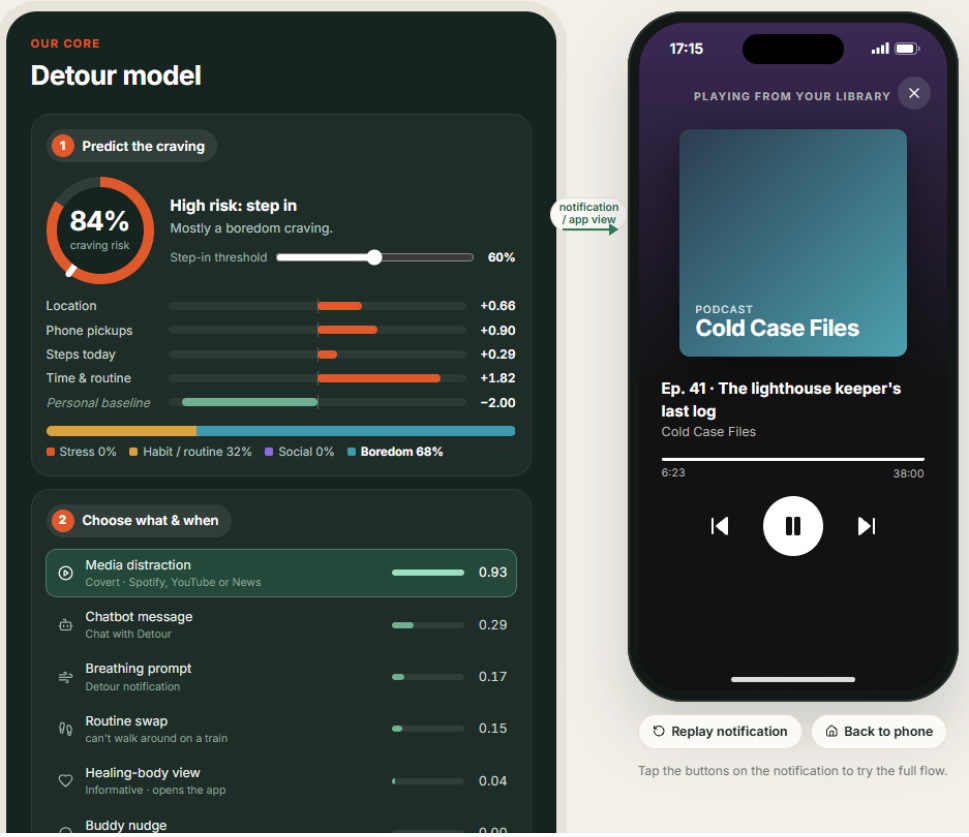


Figure 2: Enlarged view of feature selection model, and a possible feature on the user's phone, playing a song on spotify

different app, such as spotify to relax using music (figure 2).

2.5 Testing the prototype

With these 2 parts of the protoype, we could go towards possible end users for feedback. We interviewed some (ex-)smokers from our own environment, as well as non-smokers, such as peers, friends, family members. As a non-smoker could also give interesting insights into possible problems with the app interface and flow.

We proposed our prototype by pitching our idea, explaining the flow through the storyboard, while showing the user interface.

2.5.1 Received feedback

The interviewees responded positive about the following topics:

1. Smokers believed in the effectivity of Just In Time Interventions, sending notifications to help them cope with cravings, direct or indirectly related to smoking.
2. Using a chatbot is received as more pleasant than having to fill in a long questionnaire.
3. Users trying to quit smoking would be open to share usefull data, like wearable and

location information.

4. Some users like the idea of an app that learns about them and their smoking triggers, allowing them to review this data and learn more about themselves.

Meanwhile, they also identified possible problems with our app:

1. Long questionnaires and chats with chatbots are not well received and should be avoided.
2. Daily notifications, and multiple notifications throughout the day can be annoying.
3. Without notifications, some might forget to open the app.
4. Showing metrics such as craving risk could have an opposite effect: If a user sees a high risk of craving in their app, this can can be an extra trigger or excuse to actually light a cigarette.

2.5.2 Takeaways from the feedback

Presenting this visual and interactive prototype to possible users gives very usefull information about the implementation of different features, and the flow of our app. But, importantly, our app idea focuses on a framework that incorporates a large range of features, where we should be able to add (and remove) possible features, even after deployment. The user personalization is based on dynamically choosing among these features, and fitting them to the user. Therefore, due to the changing nature of our design, we will have to re-iterate the gathering of user feedback frequently to test how they perceive new features and implementations.

We can conclude that, our idea of having an app that learns about the user, and dynamically changes itself to help them in the quitting process in the best way possible, is well received and liked by the possible users. It could be a usefull and new concept, if it actually works.

This last point, raised the question "how do we make such an app actually work?", which brought us back to the drawing board. How will we connect a range of features in an intelligent way, to actually help the user to quit smoking?

We cannot just train a large model, to learn which feature will help the user quit smoking. Changing a behaviour is a complex process, requiring step by step changes, each with a reasoning behind them. The actual succes of the plan can only be seen at the end of this process.

2.6 Third prototype

In the psychological literature, there is already extensive research being done on how such behaviours, like a smoking addiction could be changed (section 3). Different models and theories are developed, for possible guidance programs. Such guidance models, form the basis of the working flow of professionals, tabacologists in our case. A tabacologist will talk to the

patient, ask questions, identify why they struggle in changing their behaviour. They will help in motivating the patient, build a plan, and help them stick to this plan.

To make our app work, we can try to learn from how it is already being done in practice, and try to fit the reasoning being done by a tabacologist, into our app. We can then try to further improve the performance by utilising the information we can gather when a larger user base is using our app.

Without going too deep on the psychological background, we will explain our reasoning behind the combination of different models.

- The HAPA model (Health Action Process Approach) identifies and classifies the person in different phases:
  - The motivational phase, where the individual forms the intention to change. Which is driven by factors such as risk perception (recognizing the health threat) and outcome expectanties (believing the change will be beneficial).
  - The volitional phase, where the individual takes action to stop. This requires factors such as action planning (planning when, where, how to act), coping planning (anticipating barriers and creating fallback strategies).
- the COM-B model, which posits that for behaviour change to occur, three interacting conditions must be met: Capability, Opportunity and Motivation. These conditions can be mapped to the different phases in the HAPA model: for example Motivation (COM-B) -> Motivational phase (HAPA)
- The Theoretical Domains Framework (TDF) acts as a magnifying glass for the COM-B model. COM-B tells us what needs to change, for example the Capability. The TDF then identifies the barriers or facilitators why people do or do not change their behaviour. For example, in the case of Capability this is:
  - Knowledge
  - Skills
  - Memory, attention and decision process
  - Behavioural regulation

When combining those three models, we can ask the user a series of questions to define: In which phase of the stopping process they are. And which barriers or facilitators (which we will call determinants) we need to target, in order to help them move forward and succesfully go through the phases.

Once we know which determinants need to change. We need to determine how we can do this using our app features.

In the psychological theory, a determinant can be changed through a Behaviour Change Technique (BCT). For example, a possible determinant can be low self-efficacy (a user doubts wether they can handle severe cravings). We can try to change this using the BCT action planning. Implementing this in a feature could be having a tool to help users build their own "if-then" coping plan.

1. To classify the possible BCTs, we can use the widely used BCT Taxonomy v1.
2. To determine which BCT alters which determinant(s), we can use the existing Theory and Techniques Tool, which provides the links showing which BCT changes which determinants.
3. We then need to identify for the possible features and interventions, which BCTs they contain. Using existing literature possibly assisted by experts in the field.

Using this, we can at each point select the feature that contains the BCTs aiming at the determinants that need to be changed in each phase.

2.7 Testing the third prototype

This is a prototype for the theoretical basis that we need to validate and improve, in order to have a logical base for our app. We interviewed a tabacologist from The Flemish Institute for Healthy Life (VIG) for feedback on this current prototype, and learned that we were on a good path, seeing clear parallels between our approach and the methods they developped at VIG. For a more detailed review, we will consult other experts in the field, like Geert Combrez, the challenger for this project who already provided a strong basis for the ideas behind the current BCT-determinant approach.

3 State of the Art

There are already hundreds of smoking cessation apps available, but unfortunately, most apps are not scientifically validated [3]. In a study by Bold et al., only 11 apps had published results supporting their efficacy.

3.1 Feasibility of apps

While many apps are available, the effectiveness of these apps may be limited [1], [6]. Luckily, it has been shown that digital interventions, through apps, can work [6], [8], when the right interventions are used. Even people who already make use of face-to-face care can find benefits in using digital help [6].

3.2 Description of existing apps

From the literature, we see that a lot of apps do not use science-based interventions [5].

To collect information about the most popular smoking cessation apps, we searched the internet for the top-rated apps, and looked at Facebook groups to see what users recommend to each other.

We looked at a few of the most popular smoking cessation apps, and we saw that they often employ these methods:

- **Progress trackers.** There are many variations of this; some track the number of days without smoking, some track the amount of money saved by not smoking, some track lifetime saved by not smoking.
- **Social aspects.** Some apps use a social aspect, like forums or other chat functions.
- **Planners.** Apps might help users plan designated smoke-free days, or plan a quit date.
- **Cognitive behavioral therapy (CBT)**. Some apps use CBT as a basis for their interventions.
- **Gamification.** Many apps use badges, mileposts, or levels to signify progress. Some apps also include games as a distraction for when a craving hits.

To make things more concrete, we will now look at a subset of the most popular apps, of which we believe they are scientifically based.

3.2.1 QuitSure

The authors of this app have published a study showing its effectiveness [2], [5]. The study reports that 80.1% of the users maintained abstinence for at least 30 days after program completion [5]. Note that this is only for users who have completed the entire program, and does not include users who dropped out, nor the people who did not respond to the follow-up questions. Also note that these are self-reported figures, and have not been biochemically verified.

The app is based on techniques from psychology, CBT, and mindfulness [5].

In an onboarding process, the user is asked to think about their smoking quantity, patterns, past quitting attempts, and relapses. The app then assesses if the user is ready to quit. After onboarding, the user is intended to follow a program consisting of content and video exercises. This program requires 6 to 10 hours over 6 to 12 days to complete [5].

In their discussion, the researchers note that there is a high dropout rate of 59.4% [5], as a result of the high engagement requirement. Additionally, most relapses were a result of high cravings, for which the app did not have specific interventions after program completion.

Unfortunately, the study does not go into detail on what specific modules in the app contributed most to successful smoking cessation.

3.2.2 Quit Sense

While not yet released to the public, there has been a pilot study that shows an 11.5% biochemically verified quit rate after 6 months [7]. This number does include the people who dropped out.

This app is a just-in-time adaptive intervention (JITAI) [7]; it tracks the users location and learns about the environment and location where the user smokes. When a user is in such an environment, the app sends in-the-moment advise to help them resist smoking.

Upon first opening the app, the user needs to set their quit date. Then, they need to submit their smoking habits and key environmental factors in real-time, via a behaviour logging tool. After the quit date, the app uses location and the previously self-identified high-risk locations to tailor the timing and contents of the support messages [7].

The app was found to be more effective than the control group that used the SmokeFree service [7] provided by the National Health Service (UK) (11.5% vs 2.9%).

The study did not find why the app worked [7], or what specific interventions helped most. Many users cited learning about their triggers and habits as being very valuable. The main reasons for dropping the app [7] were relapse, feeling like they no longer needed support, finding that the app did not match their needs, and technical issues.

3.2.3 Discussion

Both of these apps had studies showed their effectiveness in smoking cessation. However, these apps have one "expected path" to follow, and users drop the app if it doesn't fit their needs. The studies did not find why their app worked, or what specific interventions helped most.

3.3 Behavior models

Through our first meeting with Geert Crombez, we already came into contant with the Health Action Process Approach (HAPA) and Behavioral Change Techniques (BCT) models. In our own search, we came across the Capability-Opportunity-Motivation-Behavior (COM-B) and Theoretical Domains Framework (TDF) models.

These models all serve a slightly different purpose and operate on different timescales.

The COM-B model [citation needed] describes the relationship between capability, opportunity, and motivation in influencing behavior. This means you can influence behavior by tweaking the other factors. The model can thus describe what factors you need to change.

HAPA [9], [10] makes a distinction between a preintentional phase and a volitional phase. In the first phase, a person develops their motivation and plan of action. Once this "good intention" is sufficiently developed, the desired health action can be initiated. Now, in the

volitional phase, this action needs to be maintained, which needs self-regulatory skills and strategies [9].

Compared to COM-B, HAPA focuses more on the psychological processes that drive behavior change, particularly the transition from intention to action and the maintenance of that action, rather than the physical or environmental factors.

We want to use a hybrid approach, combining models to find the best intervention for the user for each stage in the program. Further research is required into how the models interact, and determine the feasibility of this idea.

3.3.1 Tools

A tool by Gezond Leven [4] shows an overview of existing BCTs. For every technique, it shows

- What it does;
- What behavior determinants it influences;
- When the technique works;
- Similar techniques;
- How to implement this technique;
- Examples;
- All references they used.

This tool will be very valuable to us when we actually want to implement these in our system.

To give an idea on what BCTs are, some examples they describe are empathy training, reinforcement, self-reflection, becoming aware of prejudices.

3.4 Related research topics

In addition to the models discussed, we also need to consider the integration of a chatbot and the use of user feedback to improve our interventions.

To successfully achieve this, we need to perform more research into the effectiveness of chatbots; when to use them, and when not to.

Additionally, we want to look into how we can effectively learn to steer our interventions when the user provides feedback. We want to explore at least two paths: how to learn from indirect feedback, like swiping away or ignoring a notification, and how to learn from direct feedback, like a yes-or-no question we ask of the user.

4 Product Vision Board

The product vision board summarises what DETOUR is, who it is for, which problem it solves and why it is better than what already exists. It is the bridge between the Design Thinking phase and the next phase: every block on the board comes from one of the earlier phases, and it sets the scope for what we will build.

4.1 Product statement

> *DETOUR is a smartphone app for smokers who want to quit, built on existing models, like the HAPA behaviour change model. HAPA determines which phase of the quit process the user is in. Within that phase, COM-B and TDF identify the determinants that need to change, such as stress, triggering places or low motivation. Wearable data (heart rate/HRV), location, agenda and check-ins help measure these determinants. DETOUR then deploys the app features whose behaviour change techniques (BCTs) target exactly those determinants, and learns per user which ones work best.*

The statement is the short version of our How Might We question from the Define phase: *"How might we choose the correct (direct vs. indirect) kind of intervention (BCT) for the right person at the correct moment to stop smoking?"* It answers the three parts of that question:

- **Right person:** personal onboarding, plus learning per user over time.
- **Right moment:** knowing the phase (HAPA), plus live context from wearable, location, agenda and check-ins.
- **Right intervention:** BCTs chosen because they target the determinant behind *this* craving.

The behaviour models are not decoration. They are the logic the app runs on. The app works as a loop:

1. User info (onboarding, check-ins, sensors) is translated into **determinants** via TDF and COM-B, within the current **HAPA** phase.
2. The determinants select the **BCTs** that target them.
3. Each BCT can be delivered through several concrete **app features**. The app picks the feature that best fits this user.
4. A **performance evaluation** checks whether the feature helped: did the user interact, did the craving pass, did they smoke? The result feeds back into the selection, which is how the app gets more **personal** over time.

Table 1 shows this for one determinant. When the app detects that social influences matter for a user (for example, the user says in onboarding that friends or a partner support their attempt), it selects BCT 3.1 *Social support (unspecified)* from the BCT Taxonomy v1 (Michie

et al., 2013). This BCT can be delivered through two different features. Which one the user gets depends on their profile, and the performance evaluation corrects the choice if it does not work.

Feature	**How it delivers social support**	**Suits users who…**	**Less suited for users who…**
Buddy system	The user links one trusted person (partner, friend, ex-smoker). When the user presses the SOS button during a craving, the buddy gets a notification and can call or send a message.	have one close person they want to involve; need help at the moment of a craving.	don't want others to know they are quitting; have no one to ask.
Leaderboard	The user joins a small group (friends or other DETOUR users) and sees each member's number of smoke-free days. The group can send each other encouragement.	are motivated by a group and by seeing progress; like a bit of competition.	feel ashamed when they fall behind; would be discouraged after a lapse.

Table 1: One BCT, two features. Determinant: social influences (TDF), which falls under social opportunity in COM-B. BCT: 3.1 Social support (unspecified). Each row is a separate feature that delivers the same BCT. The two right columns show which user profile the app uses to choose between them. If the chosen feature is ignored or does not reduce cravings, the performance evaluation lowers its score for that user and the other feature is tried.

4.2 Target group

Smokers who want to quit. This covers three groups that we met during the Empathize phase:

- people who tried before and failed,
- people who have stopped and want to keep it that way (relapse prevention),
- people who are trying for the first time.

All three share the condition we heard in every interview and in the first meeting with our challenger, Prof. Geert Crombez: **the motivation has to come from the user.** DETOUR does not try to convince people who don't want to quit; it helps people who already do.

4.3 Needs

We identified two core needs:

1. **Personal fit.** Smokers don't know which quit method or app suits them. There are many apps, most of low quality, and testing them one by one costs time and motivation. In practice, a failed attempt with the wrong app feels like a personal failure.
2. **Craving support that matches the cause.** Cravings have different causes (stress, a place, boredom, a social situation, physical withdrawal), and the same technique does not work for every cause. The HAPA phase also matters: someone still building motivation needs different help from someone resisting a craving on day three.

These needs come directly from the interviews (Empathize) and from the clustering of our HMW questions (Define), where "the right method for the right person" and "help at the moment of the craving" were the two largest clusters.

4.4 Product

- **Personal onboarding.** The user maps their own triggers and preferred kinds of help. Based on user feedback from the Test phase, this happens through a short chatbot conversation instead of a questionnaire.
- **Smart selection from a BCT library.** The app picks the techniques, not the user. Selection works on two levels:
  - **Phase (HAPA):** before the quit day, only motivation modules (health risks, money wasted, confidence). After the quit day, only coping modules (coping plans, trigger management, streaks).
  - **Cause (COM-B):** stress gets urge-surfing audio, a trigger place gets a problem-solving chatbot, low motivation gets the user's personal "why".
- **Several features per BCT.** The same technique can be delivered in different ways (Table 1), so the app can match the form to the person.
- **Learning per user.** Through the performance evaluation, the app learns which features work for this user and which ones they ignore.

4.5 Value

- **No need to compare and test apps yourself.** DETOUR picks the right techniques for you.
- **Help that fits the cause of this craving and the phase you're in**, so less wasted effort and a higher chance of quitting for good.

Users should not fail their quit attempt because the method didn't suit them.

4.6 Competitors and alternatives

Most quit apps (e.g. QuitNow, Kwit, Smokefree, Stoppen met Roken) do one of two things: they track progress and savings, or they offer the same fixed step-by-step programme to everyone. A few are science-based (Quit Genius, QuitSure, WHO Quit Tobacco), but they also follow a fixed path.

Apps that adapt to the user's context in real time exist only in research, for example Q Sense (geofence-triggered support), Smart-T (EMA-based lapse prediction) and Sense2Stop (stress detection from wearable sensors). These prove the idea is feasible, but none is on the market for consumers. DETOUR fills that gap by combining the scientific basis of these research apps with the usability of consumer apps. See the State of the Art chapter for the full comparison.

4.7 Link with the recommendations

The vision board describes *what* we will build. The recommendations from the Test phase set limits on *how* (Table 2).

Recommendation	**Effect on the product**
Keep actions concrete	Every BCT module ends in one clear action, e.g. "do this 3-minute exercise", not "try to relax"
Long questionnaires don't work; a chatbot works better	Onboarding and check-ins are short chatbot conversations
Long chats are not pleasant either	Chatbot conversations are capped at a few turns
Showing the craving level can backfire	Risk is used internally to time interventions, but is not shown to the user as a score
Some users dislike daily notifications	Notification frequency is part of the personal profile and is learned over time
Adapt language to the profile	Tone (formal, informal, technical) is set during onboarding

Table 2: Recommendations from the Test phase and how they change the product described on the vision board.

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A Appendix: Your first appendix

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B Appendix: Your second appendix

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